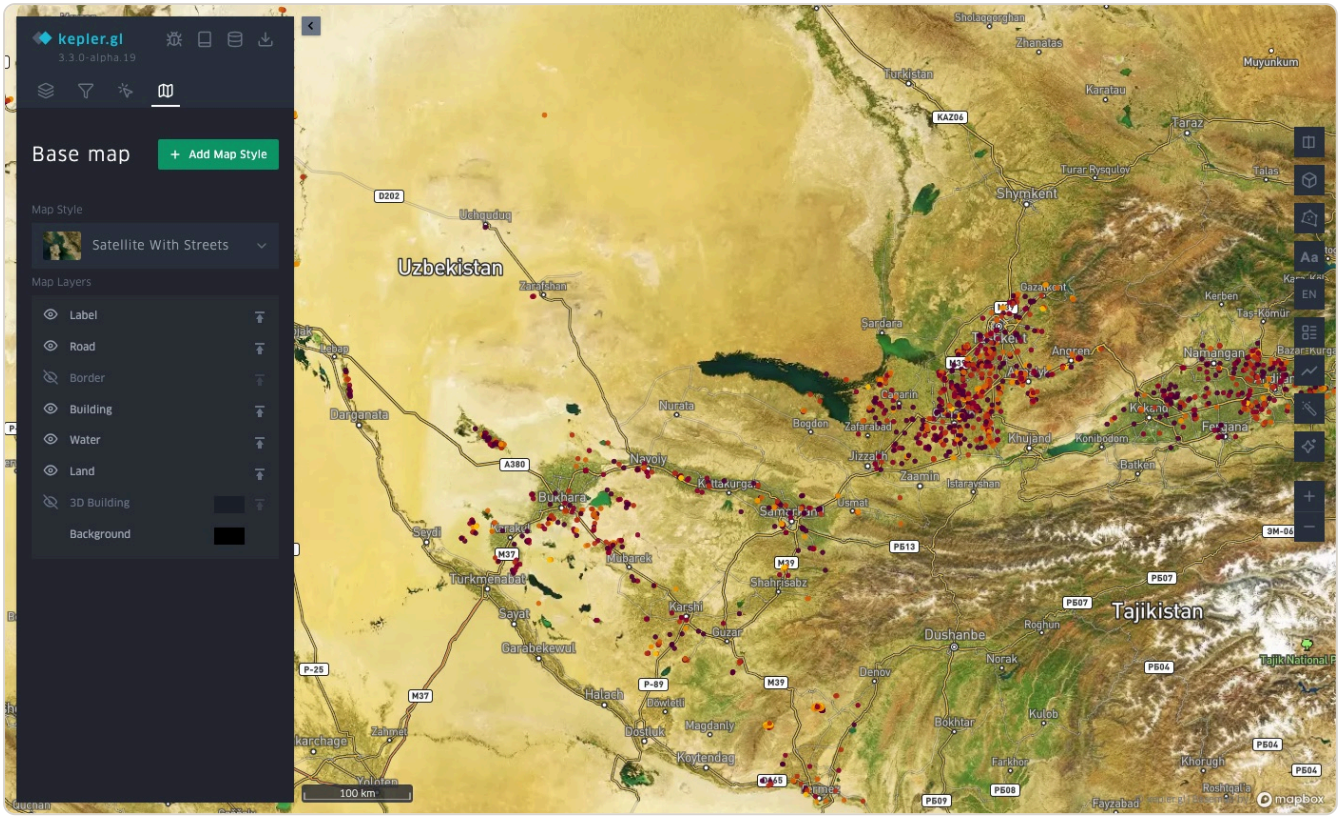
Day 01 · Points – Fires Seen From Space 🛰️🔥

In this tutorial you'll make a map of every fire that NASA satellites spotted in Uzbekistan in 2025, **without installing anything and without writing any code**. Everyone starts with the same data, but you'll each pick a different way to show it, so every map in our group comes out different.

⏱️ Time	💻 You need	🧰 Tool
30–45 minutes	A computer with Chrome, Edge or Firefox	kepler.gl (free, runs in the browser)



📄 **Printable PDF:** English · Русский · O‘zbekcha

🛰️ Part 0 – Meet the data: NASA FIRMS fire detections

Where does it come from? Two NASA satellites, **Terra** and **Aqua**, orbit the Earth about 700 km up. Each one carries a camera-like sensor called **MODIS**. MODIS doesn't only take normal photos. It also measures **heat** (thermal infrared).

How does a satellite "see" a fire? MODIS splits the ground into squares of about **1 km × 1 km** (called *pixels*). When one square is much hotter than the squares around it, NASA's computers mark it as a **hotspot**. Each hotspot becomes **one point** in our file, placed at the centre of that 1 km square.

> 💡 So a point means "*something was burning somewhere inside this 1 km square at that moment*". It isn't the exact spot of the fire.

When does it look? Between them, the two satellites pass over Uzbekistan about **4 times a day**: Terra around 10:30 and 22:30, Aqua around 13:30 and 01:30. That's why the data has both **day** and **night** detections.

Not every hotspot is a wildfire. Gas flares at oil and gas fields and some factories are hot all the time. NASA labels these as *static sources*. In our file they're called **"Gas flare / industry"**.

This system is called FIRMS (Fire Information for Resource Management System). Firefighters, farmers and scientists around the world use it every day.

What's inside our file

We prepared the 2025 data for Uzbekistan as one table: `data/fires_uzbekistan_2025.csv`. Each row is one point (**4,649 points** in total).

Column	What it means	Example
`latitude`, `longitude`	Where the point is	41.25, 69.21
`date`	Day it was detected	2025-03-14
`region`	Which viloyat it falls in	Surkhandarya
`fire_type`	Vegetation fire, or gas flare / industry	Vegetation fire
`day_night`	Seen during the day or at night	Day
`season`	Winter / Spring / Summer / Autumn	Spring
`month`	Month of the year	03 Mar
`satellite`	Which satellite saw it	Terra or Aqua
`confidence_level`	How sure NASA is that it's a real fire	Low / Nominal / High
`fire_power_mw`	How strong the heat was, in megawatts	21.3

A few facts to get you curious:

- 🌾 **March** had the most detections of any month (924). Many farmers burn fields in early spring.
- 🏭 **2,000 of the 4,649 points** are gas flares or industry, not wildfires.
- 🌙 About **1 in 3** detections happened **at night**.

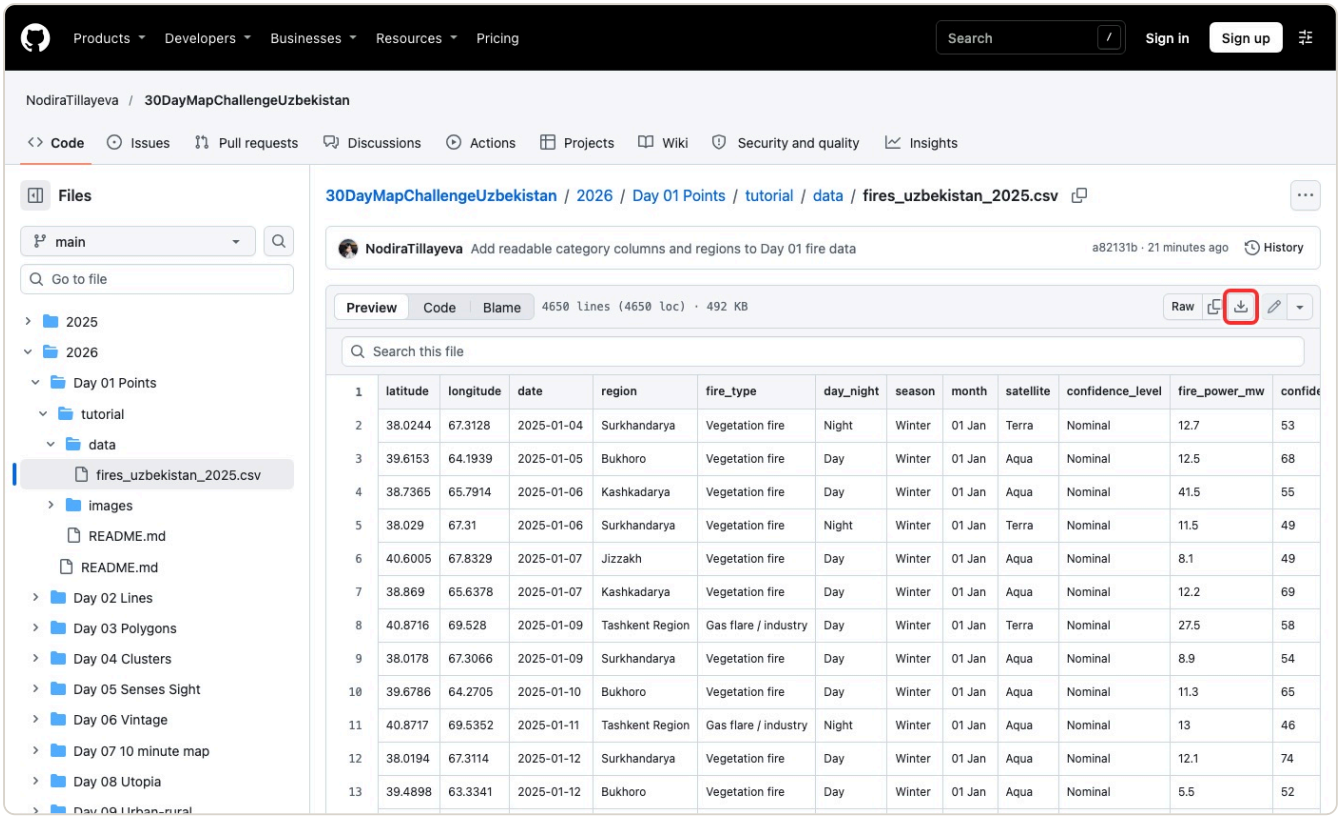
Part 1 – Download the data

Download the data to your computer first. You have two options, and both are free with no sign-up.

	Option A – Our ready-made file	Option B – Straight from NASA
Best for	Your first time ✅	Trying another year or country
Columns	Easy words: *Day/Night*, *Spring*, *Vegetation fire*, region names	NASA's short codes: `D` / `N`, `0` / `2`, numbers

Option A – Download our ready-made file from GitHub

Step 1. Open **fires_uzbekistan_2025.csv**. GitHub shows the data as a table. Click the **download icon** ⬇️ above the table, on the right. The file `fires_uzbekistan_2025.csv` is saved to your **Downloads** folder.

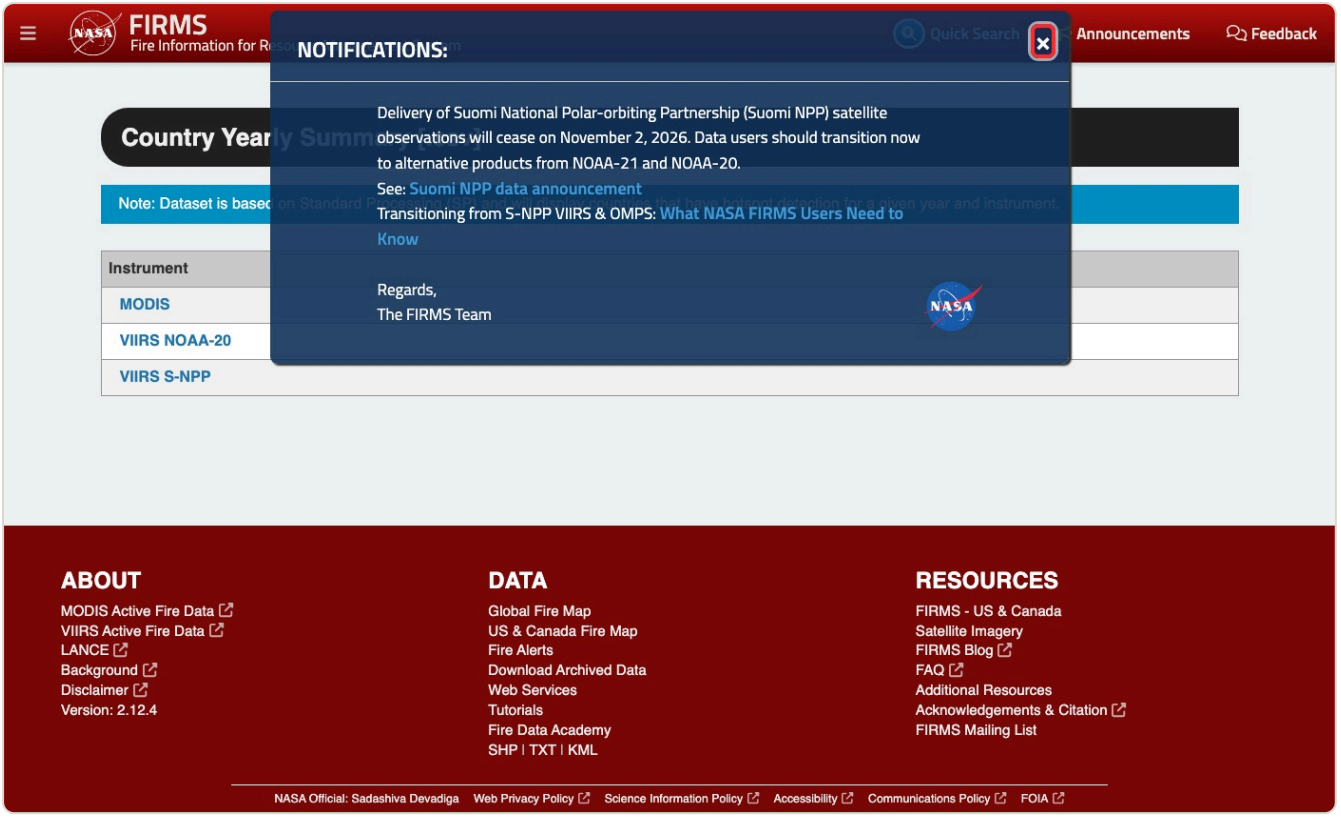


➡️ Now go to **Part 2**.

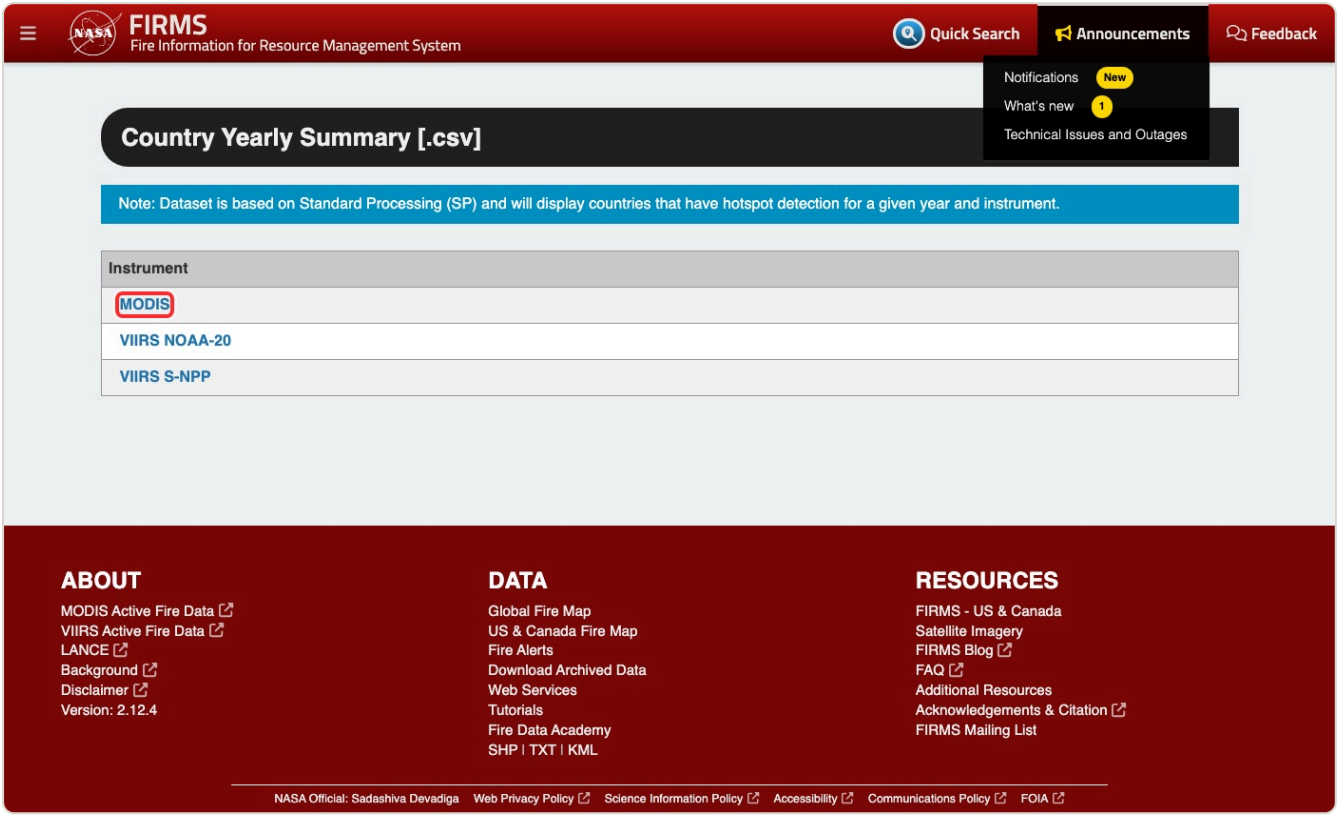
Option B – Download straight from NASA FIRMS

This is where the data really comes from. Once you know these four clicks, you can get fire data for **any country, from 2000 to today**.

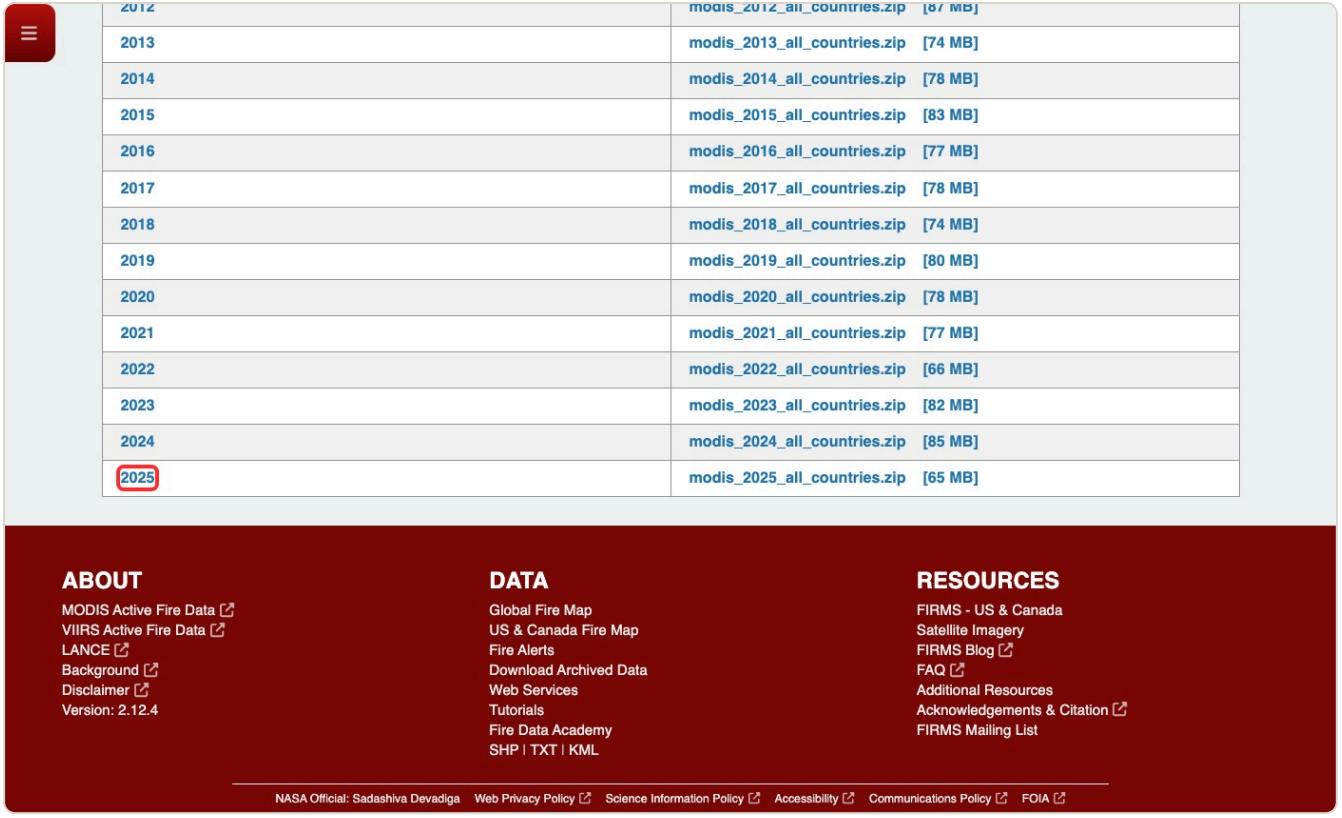
Step 1a. Open **firms.modaps.eosdis.nasa.gov/country**. If a blue **NOTIFICATIONS** box appears, close it with the ×.



Step 1b. Under **Instrument**, click **MODIS**. That's the sensor on the Terra and Aqua satellites.

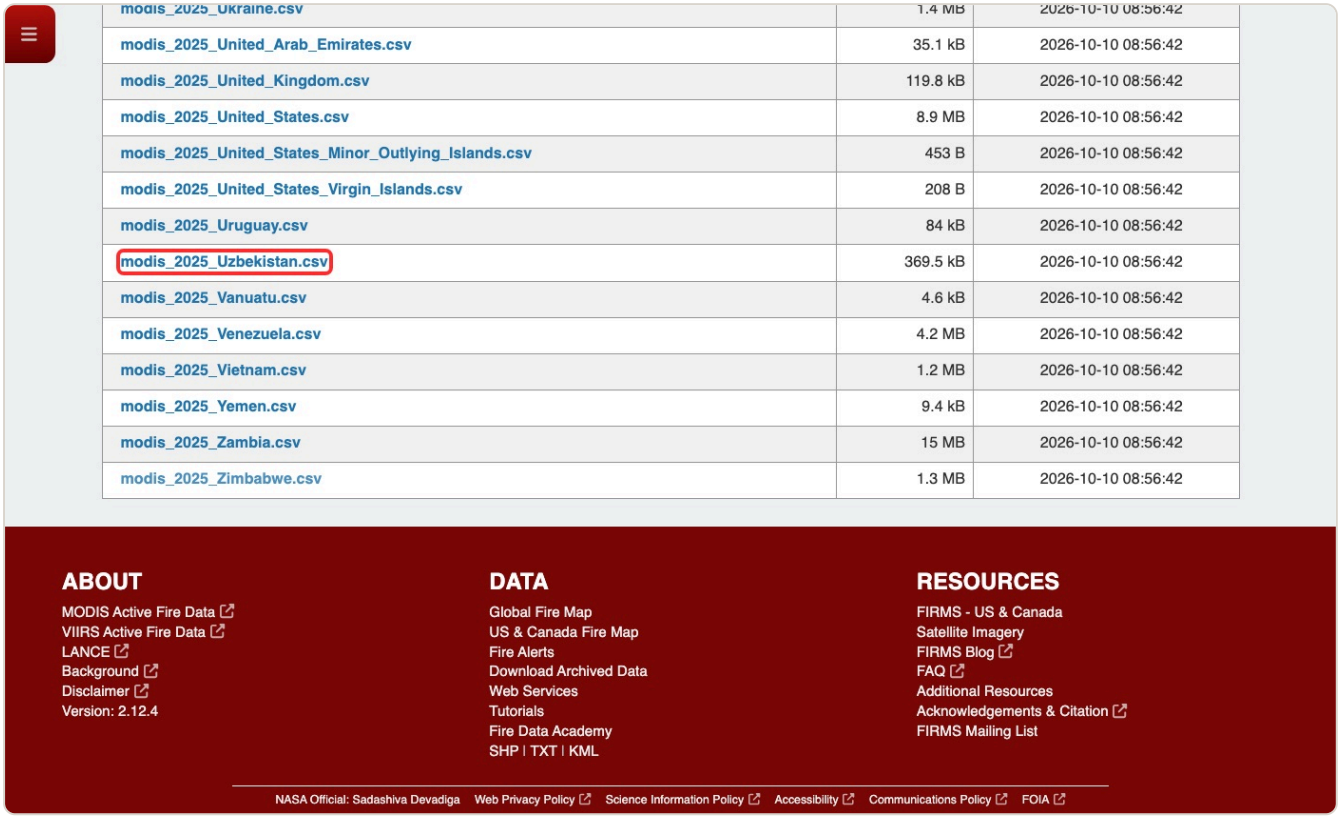


Step 1c. Scroll down and click the **year** you want, e.g. **2025**. Click the year number on the **left**, not the big `.zip` file. The zip has every country in the world and is huge.



2012	modis_2012_all_countries.zip [87 MB]
2013	modis_2013_all_countries.zip [74 MB]
2014	modis_2014_all_countries.zip [78 MB]
2015	modis_2015_all_countries.zip [83 MB]
2016	modis_2016_all_countries.zip [77 MB]
2017	modis_2017_all_countries.zip [78 MB]
2018	modis_2018_all_countries.zip [74 MB]
2019	modis_2019_all_countries.zip [80 MB]
2020	modis_2020_all_countries.zip [78 MB]
2021	modis_2021_all_countries.zip [77 MB]
2022	modis_2022_all_countries.zip [66 MB]
2023	modis_2023_all_countries.zip [82 MB]
2024	modis_2024_all_countries.zip [85 MB]
2025	modis_2025_all_countries.zip [65 MB]

Step 1d. Scroll down to your country and click it, e.g. **modis_2025_Uzbekistan.csv** (about 370 kB). The file is saved to your **Downloads** folder.



modis_2025_Ukraine.csv	1.4 MB	2026-10-10 08:56:42
modis_2025_United_Arab_Emirates.csv	35.1 kB	2026-10-10 08:56:42
modis_2025_United_Kingdom.csv	119.8 kB	2026-10-10 08:56:42
modis_2025_United_States.csv	8.9 MB	2026-10-10 08:56:42
modis_2025_United_States_Minor_Outlying_Islands.csv	453 B	2026-10-10 08:56:42
modis_2025_United_States_Virgin_Islands.csv	208 B	2026-10-10 08:56:42
modis_2025_Uruguay.csv	84 kB	2026-10-10 08:56:42
modis_2025_Uzbekistan.csv	369.5 kB	2026-10-10 08:56:42
modis_2025_Vanuatu.csv	4.6 kB	2026-10-10 08:56:42
modis_2025_Venezuela.csv	4.2 MB	2026-10-10 08:56:42
modis_2025_Vietnam.csv	1.2 MB	2026-10-10 08:56:42
modis_2025_Yemen.csv	9.4 kB	2026-10-10 08:56:42
modis_2025_Zambia.csv	15 MB	2026-10-10 08:56:42
modis_2025_Zimbabwe.csv	1.3 MB	2026-10-10 08:56:42

> 💡 **Tip:** press **Ctrl + F** (or **Cmd + F** on a Mac) and type the country name to find it quickly in the long list.

📖 How to read NASA's columns (Option B only)

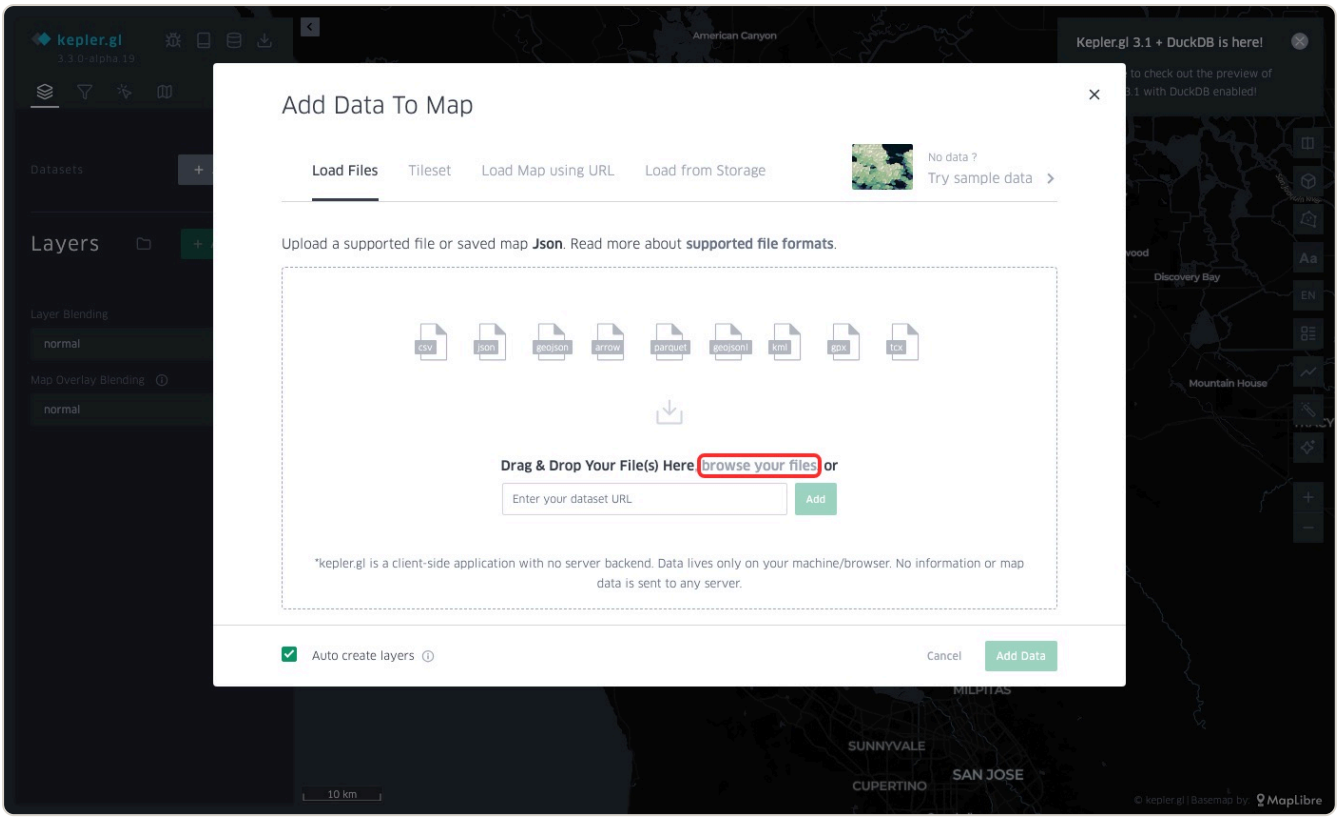
NASA's file uses short codes. Here's what they mean, and which easy-word column in our ready-made file matches each one:

NASA column	What it means	In our file
`acq_date`	Date of detection	`date`
`acq_time`	Time in **UTC**, written as HHMM (add 5 hours for Uzbekistan time)	`acq_time_utc`
`daynight`	`D` = day, `N` = night	`day_night`
`type`	`0` = vegetation fire, `1` = volcano, `2` = other static heat source (gas flares, factories), `3` = offshore	`fire_type`
`confidence`	0–100, how sure NASA is that it's a real fire	`confidence_level` (Low < 30, Nominal 30–80, High > 80)
`frp`	Fire Radiative Power: how strong the heat was, in megawatts	`fire_power_mw`
`satellite`	`Terra` or `Aqua`	`satellite`
`brightness`	Temperature of the pixel in Kelvin	`brightness_k`

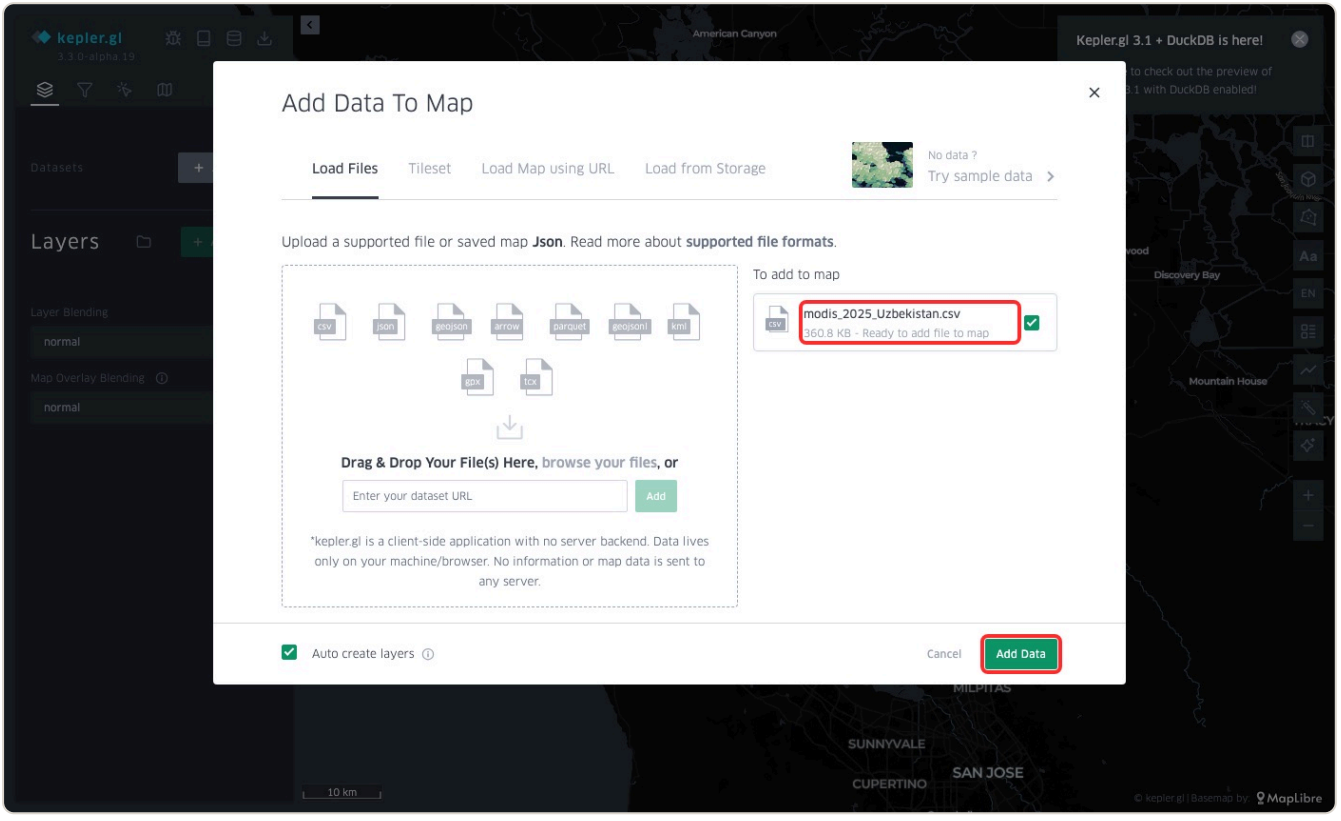
NASA's file has **no** `region`, `season` or `month` columns. We added those. With NASA's file, the easiest columns to color by are **`daynight`** and **`satellite`**. `confidence` and `frp` give a smooth color gradient.

Part 2 – Load the data into kepler.gl

Step 2. Open **kepler.gl/demo**. An **"Add Data To Map"** window opens by itself. **Drag your downloaded file** from the Downloads folder into the dotted box, or click **browse your files** and choose it.



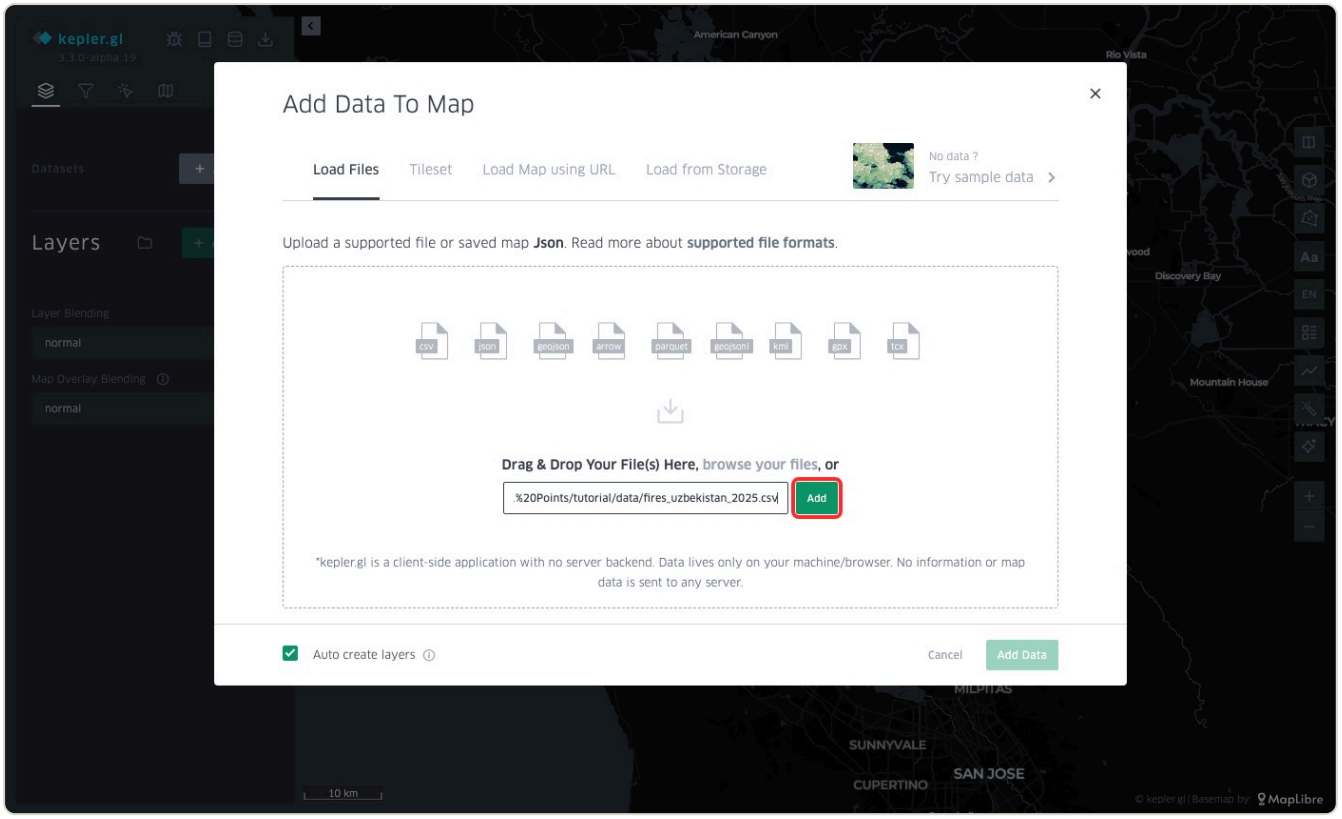
Step 3. The file appears on the right. Click the green **Add Data** button.



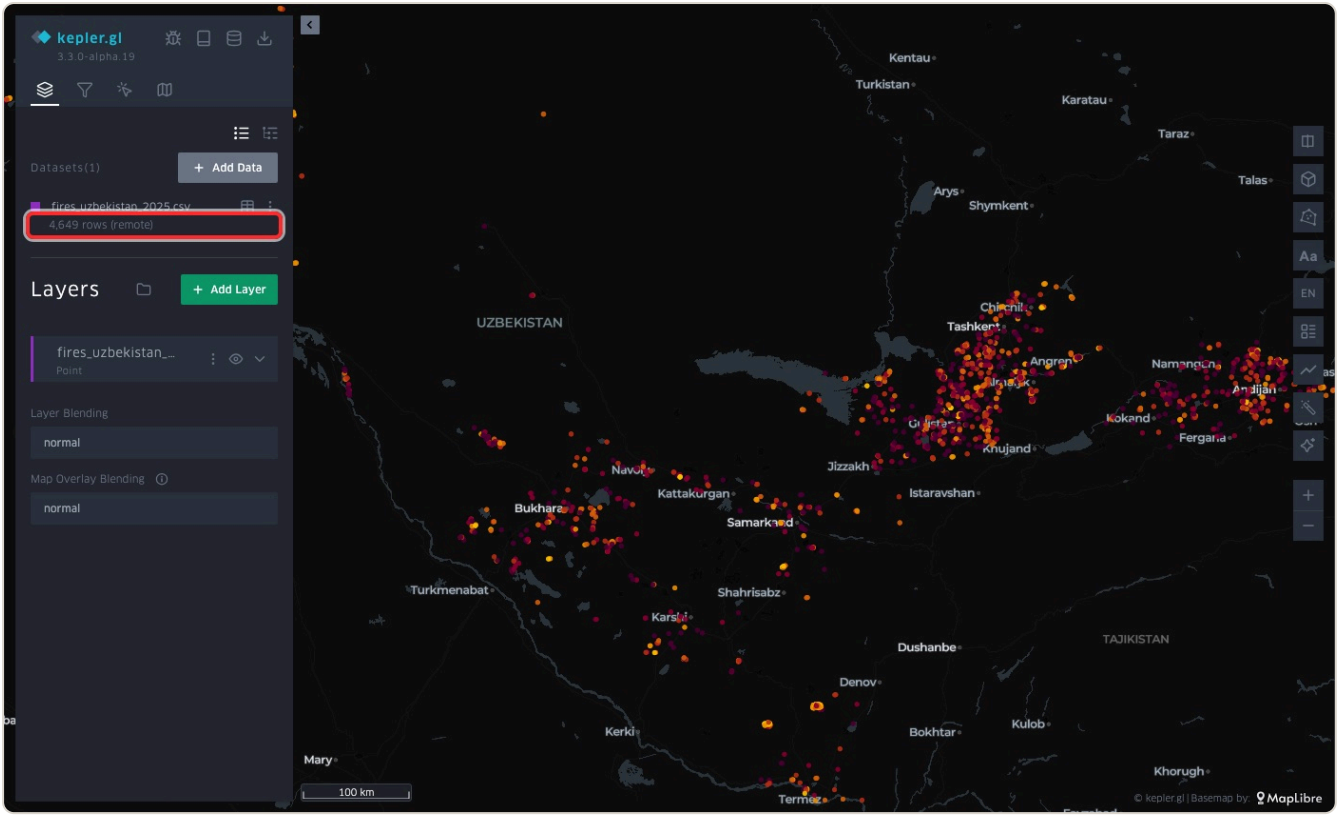
⚡ Shortcut: load our file without downloading (paste a link)

Instead of a file, you can paste this link into the **"Enter your dataset URL"** box and click **Add**, then **Add Data**:

```
https://raw.githubusercontent.com/NodiraTillayeva/30DayMapChallengeUzbekistan/main/2026/Day%2001%20Points/tutorial/data/fires_uzbekistan_2025.csv
```



Step 4. 🎉 **Your points are on the map!** Each dot is one fire seen from space. The panel on the left says **4,649 rows**, one for each point.

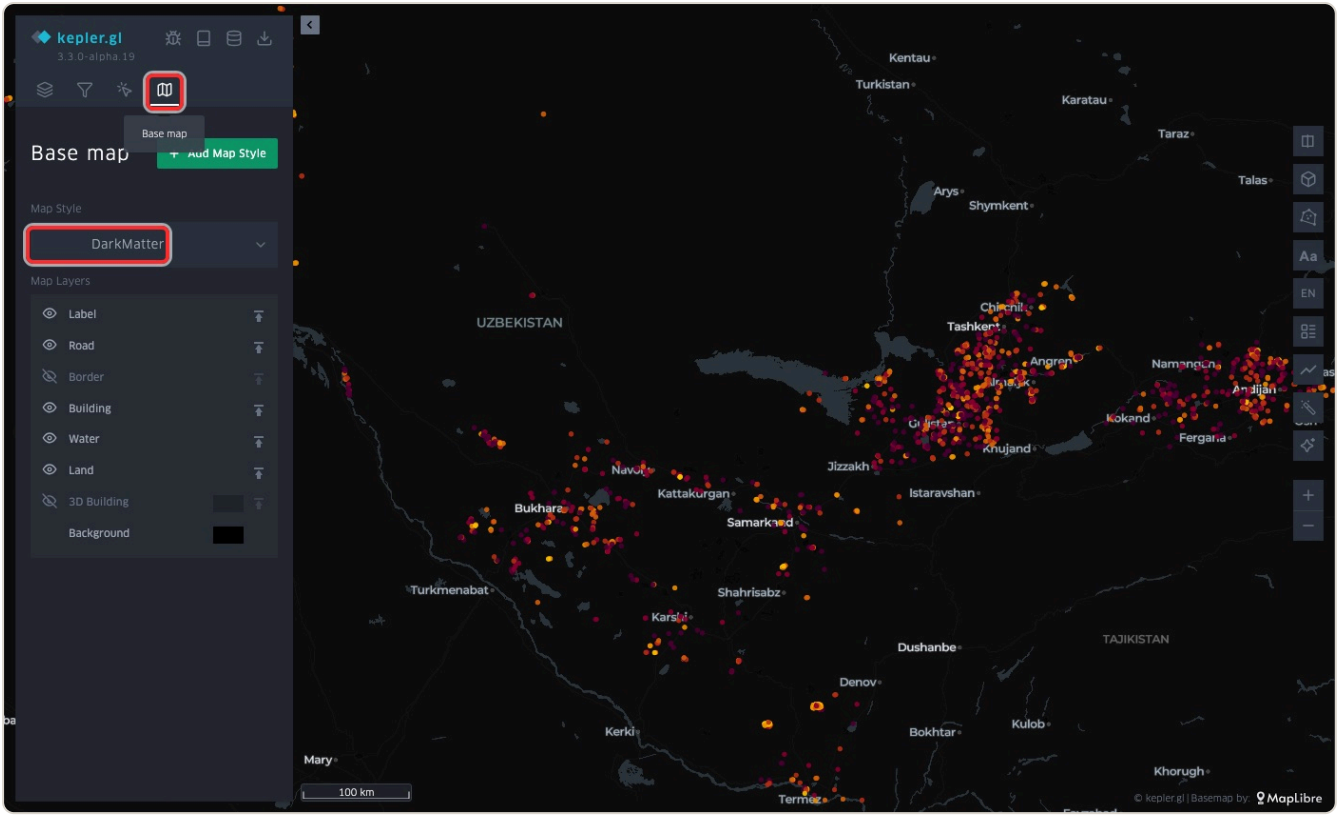


Moving around: drag the map to move it, scroll your mouse wheel to zoom in and out, and hold **right-click** while dragging to tilt or rotate.

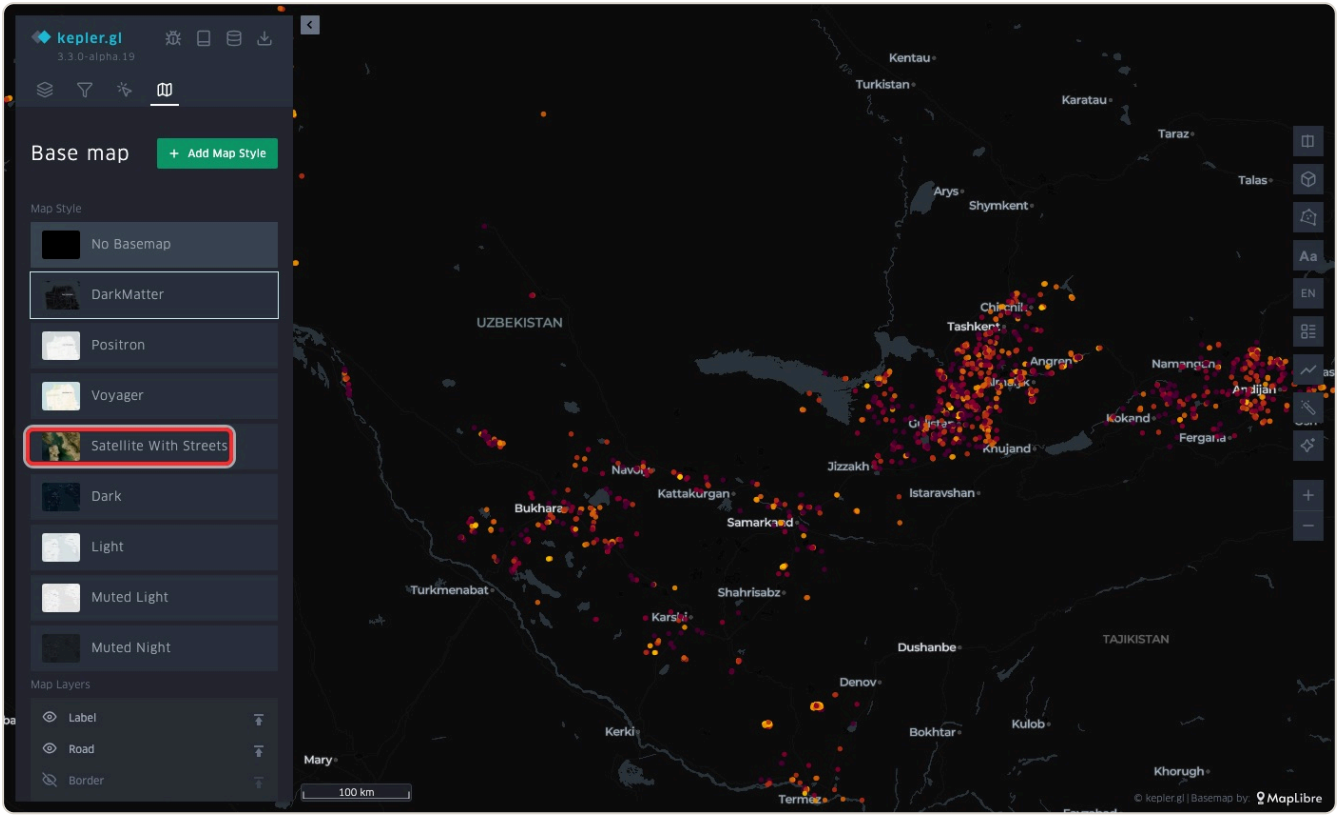
❌ If a pop-up about "DuckDB" or a survey shows up, close it with its ×. Otherwise it will appear in your final image.

Part 3 – Put a satellite photo underneath

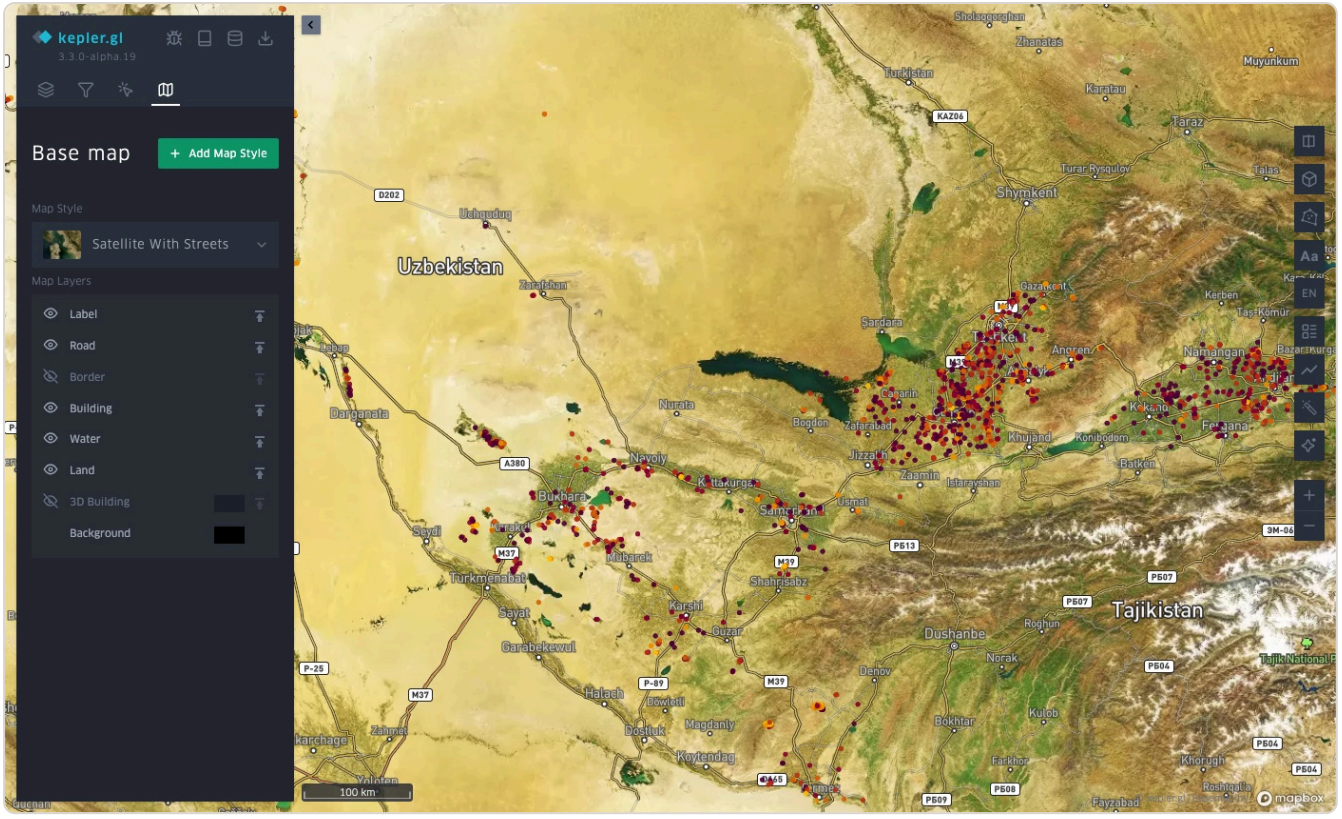
Step 5. Click the **map icon** (the 4th tab at the top of the left panel). Then click the **Map Style** box, which says *DarkMatter*.



Step 6. Choose **Satellite With Streets**.



Step 7. Now you can see the deserts, mountains, rivers and green farmland of Uzbekistan under your points. Notice how most fires sit on the **green farmland**, not in the desert.

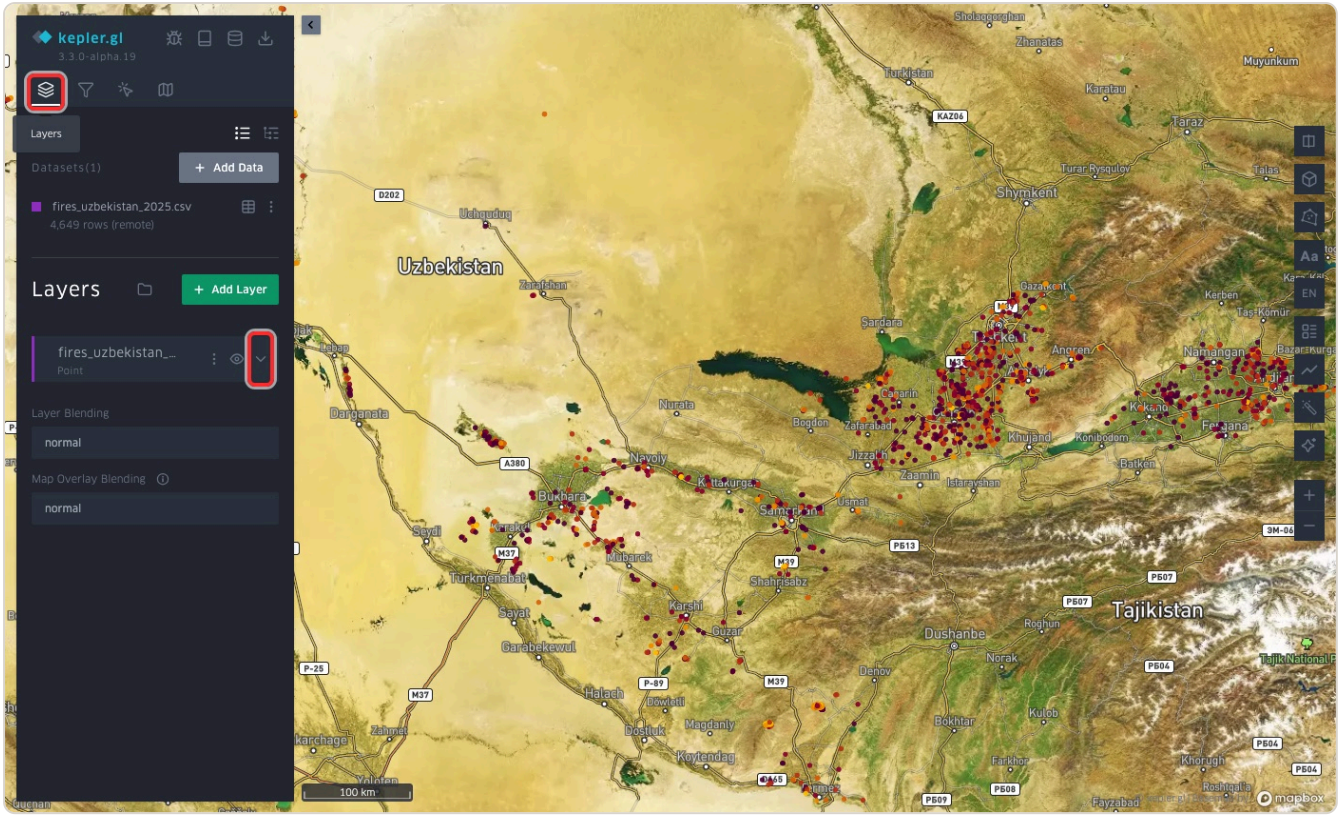


🎨 Not a fan of the satellite look? *Dark*, *Light* or *Voyager* also look great. Choosing a different background is an easy way to make your map yours.

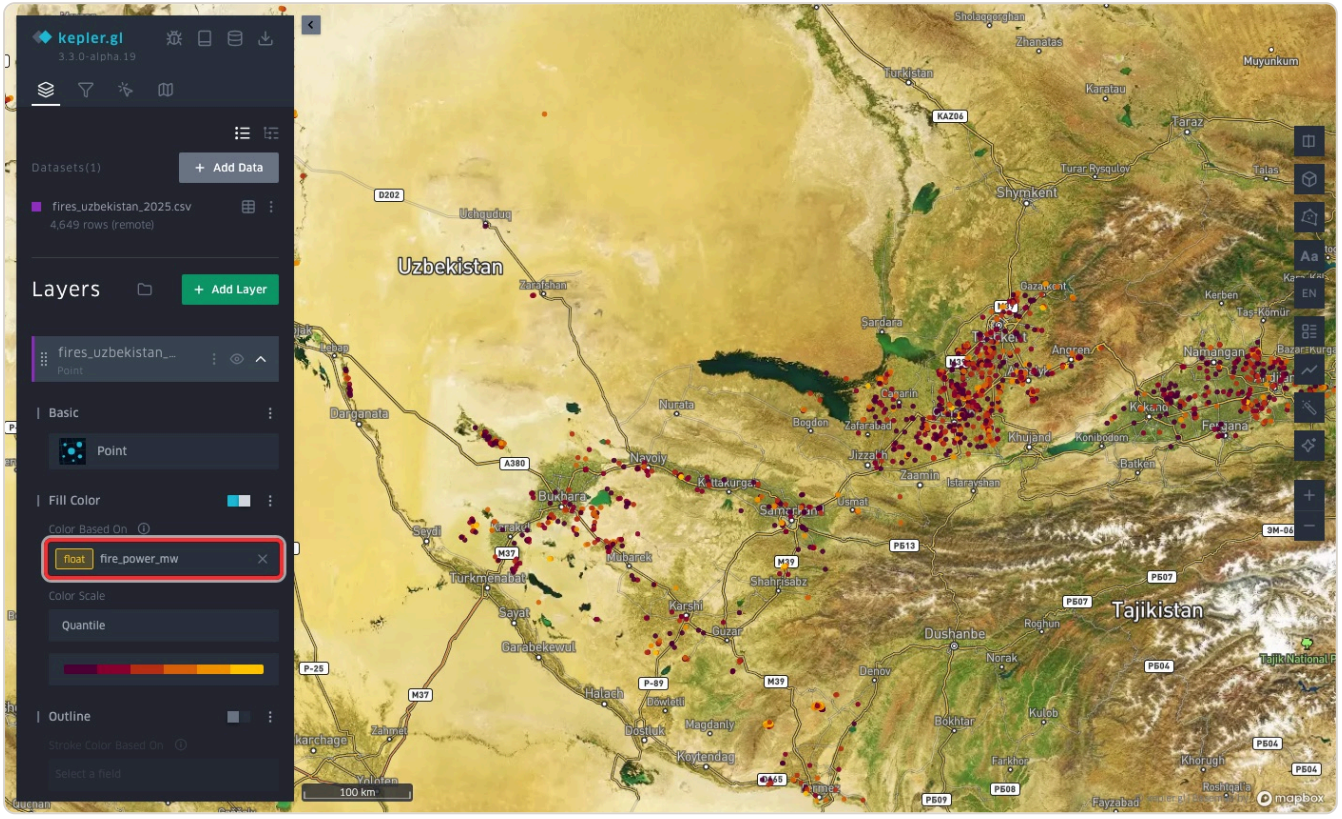
Part 4 – Color the points by category

Right now the points are colored by brightness. Let's color them by something more meaningful.

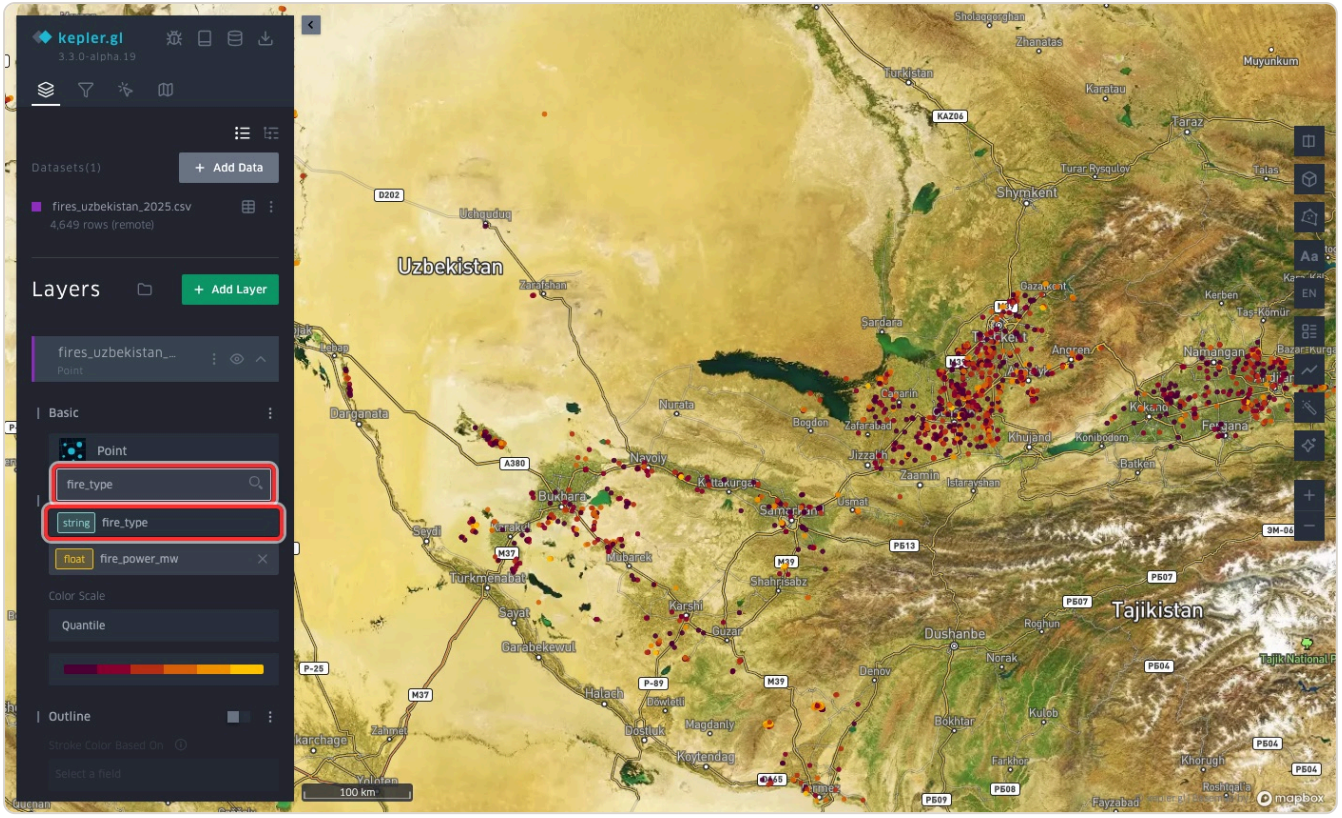
Step 8. Go back to the **layers icon** (the 1st tab). Click the small **arrow** ⌄ next to the layer name to open its settings.



Step 9. Under **Fill Color**, click the box under **Color Based On**.

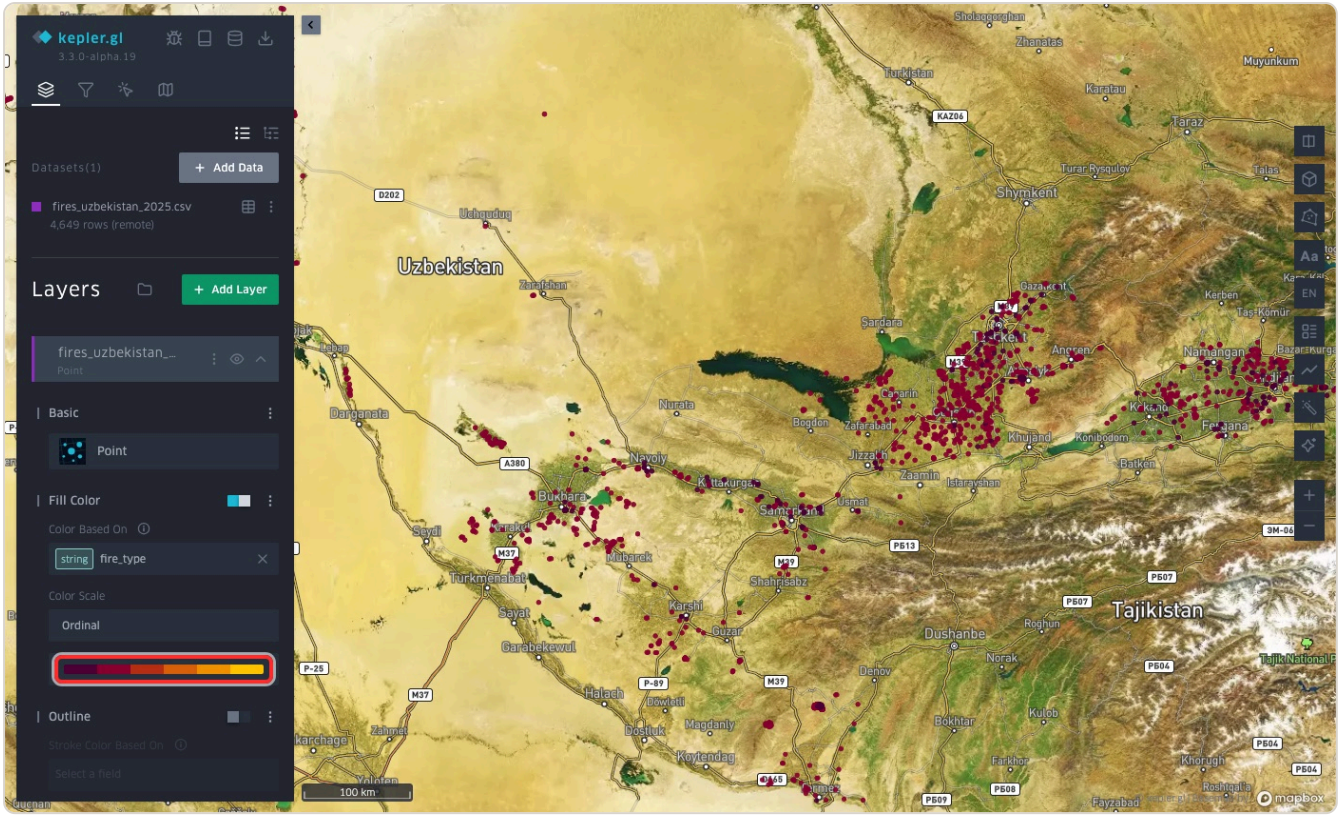


Step 10. Type the name of a column in the search box. In this example we use `fire_type`. Click it in the list.

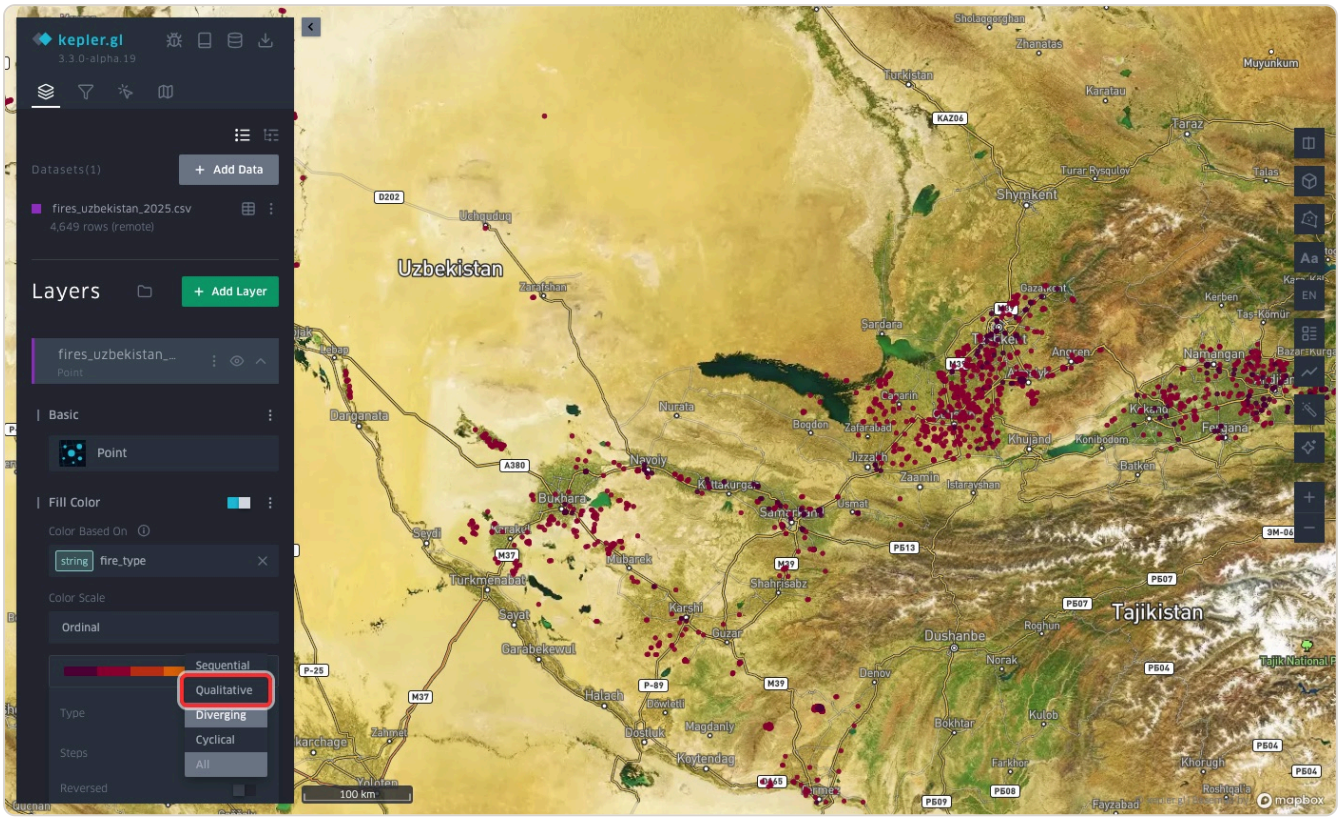


👉 **This is the step where your map becomes different from everyone else's.** See **Part 6** and pick your own column.

Step 11. The two categories may get colors that look almost the same. To fix this, click the **colored bar** under *Color Scale*.



Step 12. Next to **Type**, click **All** and choose **Qualitative**. Qualitative palettes give every category a clearly different color.



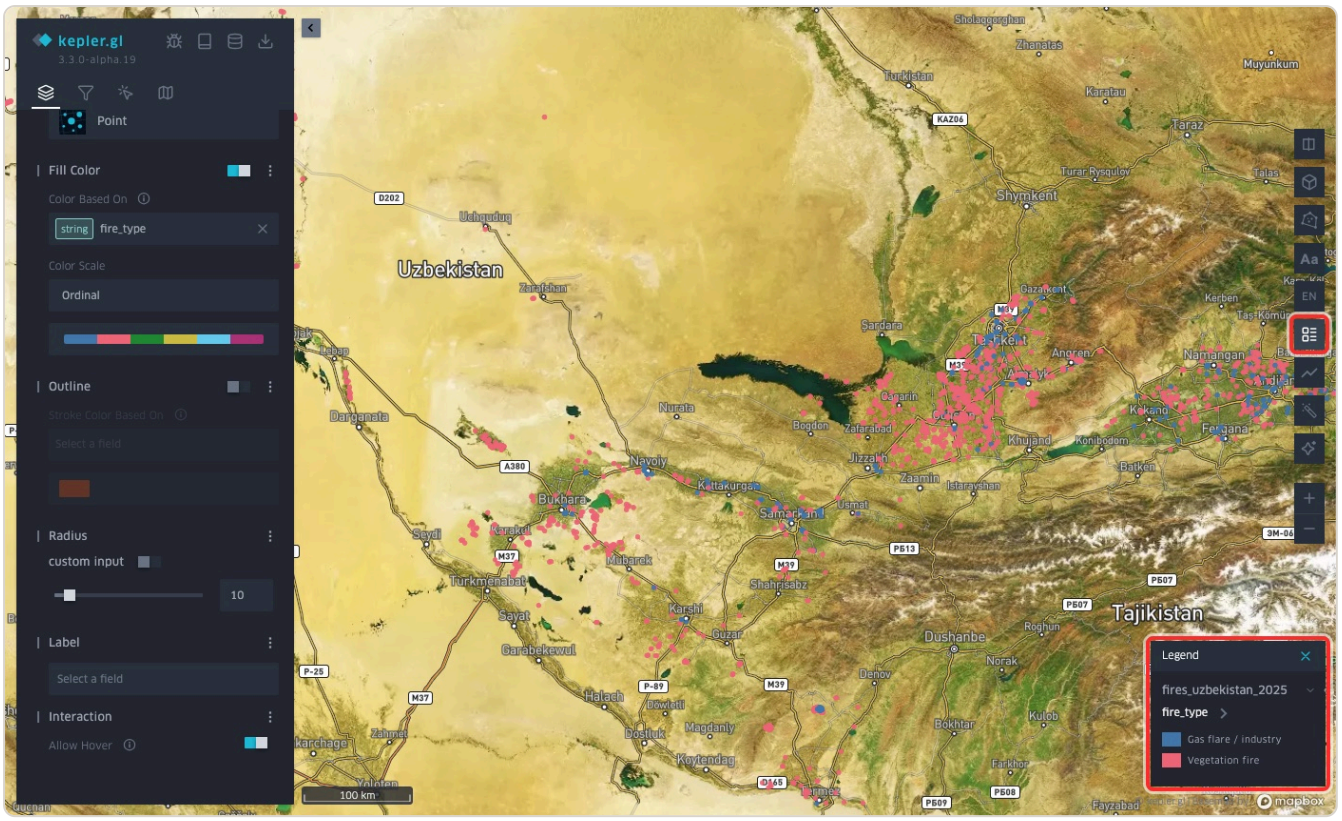
Step 13. Click any palette you like in the list.



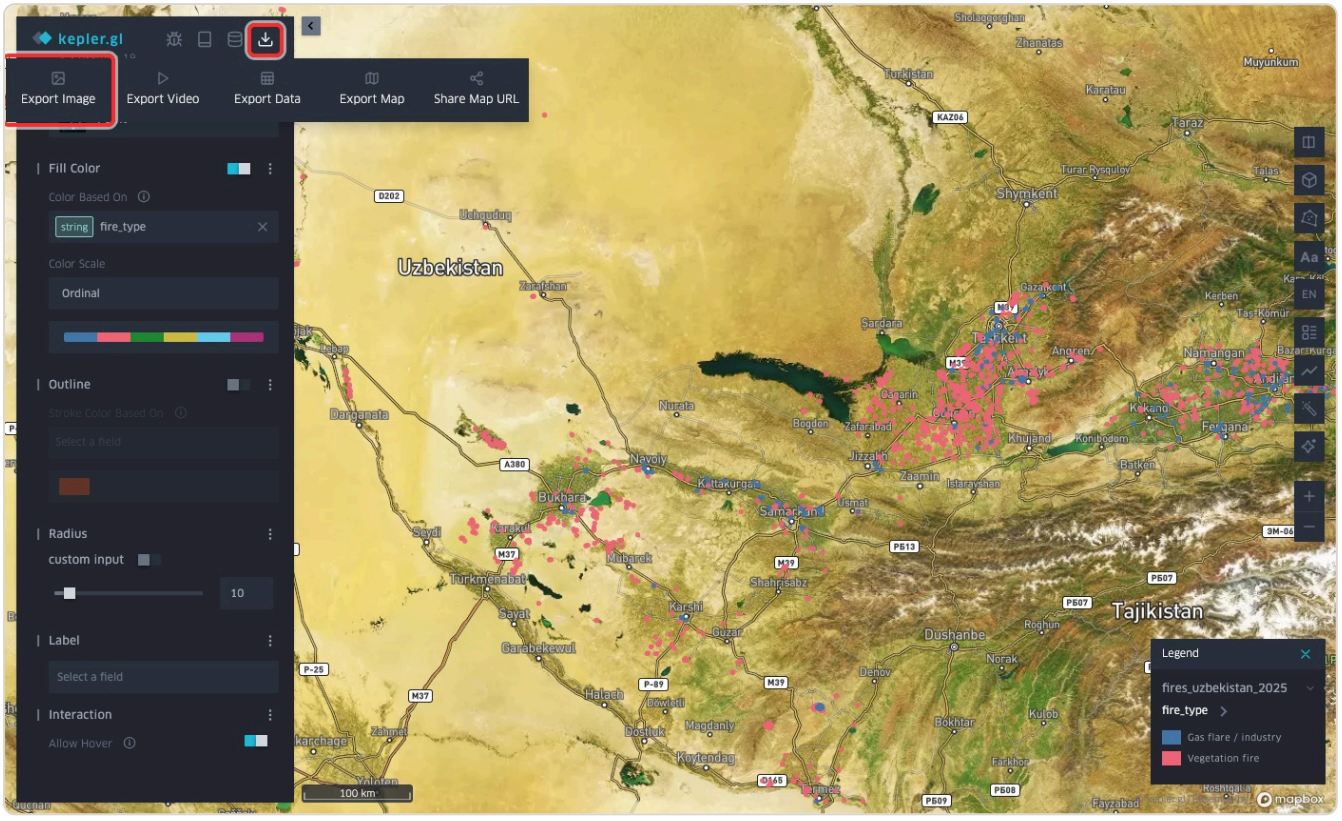
🔢 **Lots of categories?** (`region` has 14!) Change **Steps** from 6 to **12** so the colors don't repeat.

Part 5 – Add a legend and save your map as an image

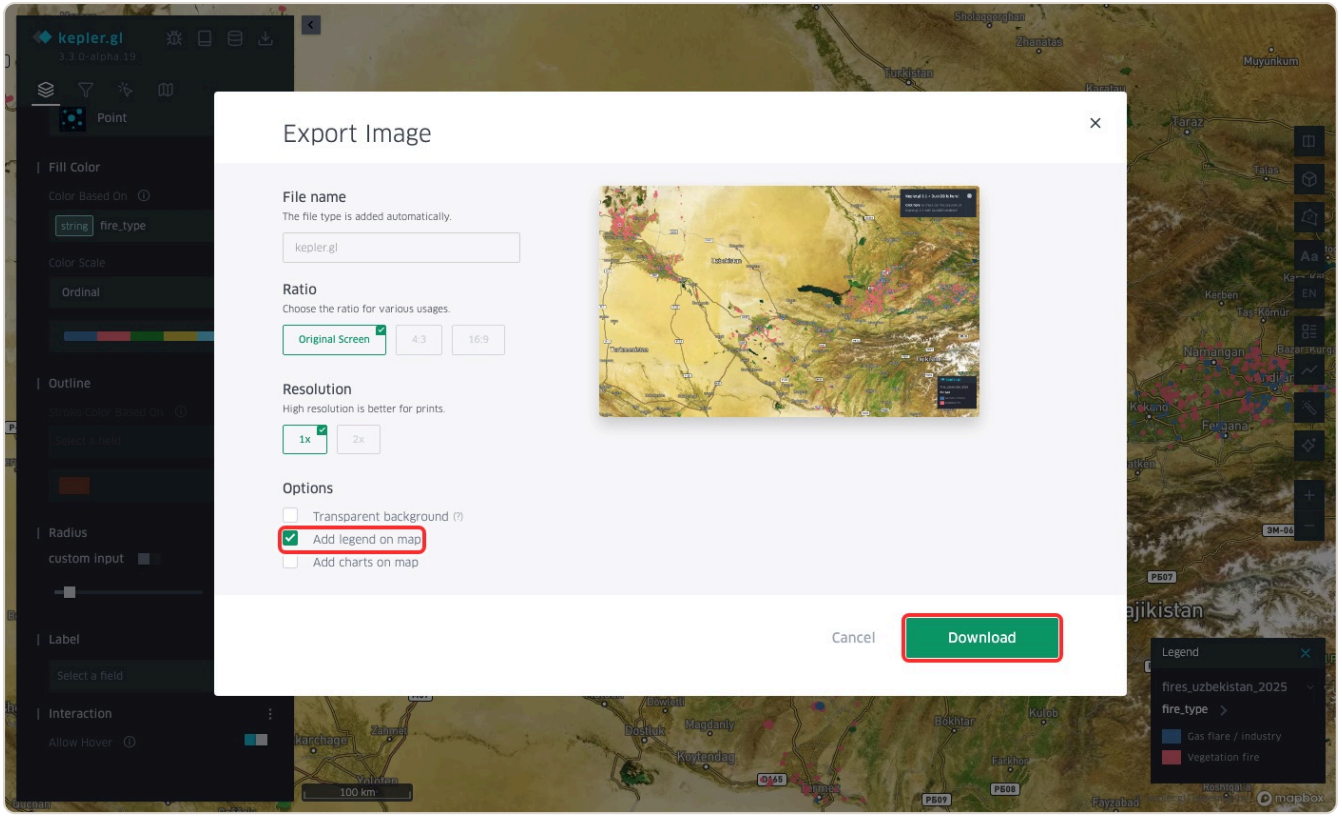
Step 14. Click the **legend button** on the right side of the map. A small box now explains your colors. *Gas flare / industry* and *Vegetation fire* now show up as clearly different colors.



Step 15. Zoom and move the map until it looks the way you want. Then click the **download icon** (top of the left panel) and choose **Export Image**.



Step 16. Tick **Add legend on map**. Under **Resolution**, choose **2x** if you'd like a sharper image. Wait for the preview to appear, then click **Download**. A `.png` file is saved to your **Downloads** folder.



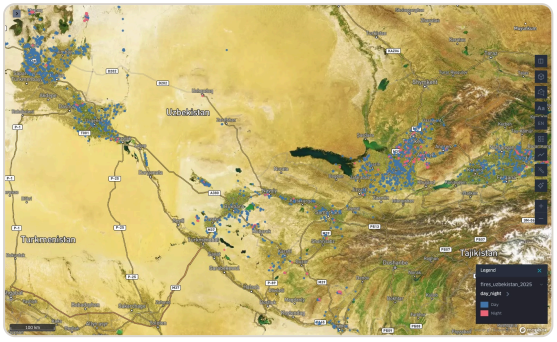
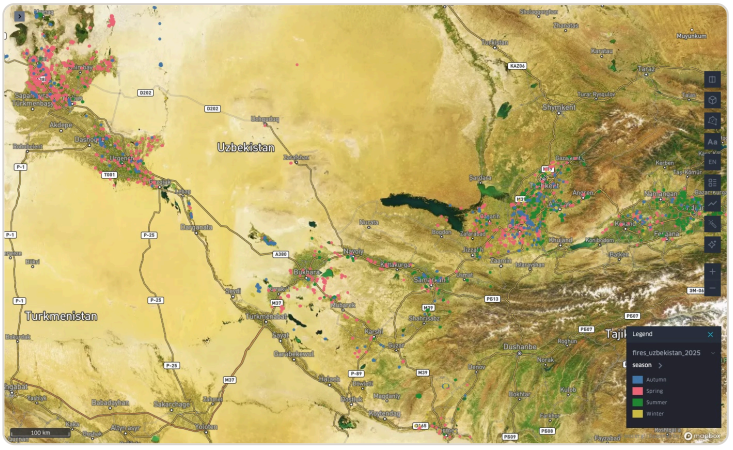
🎉 **You've made a map from satellite data!**

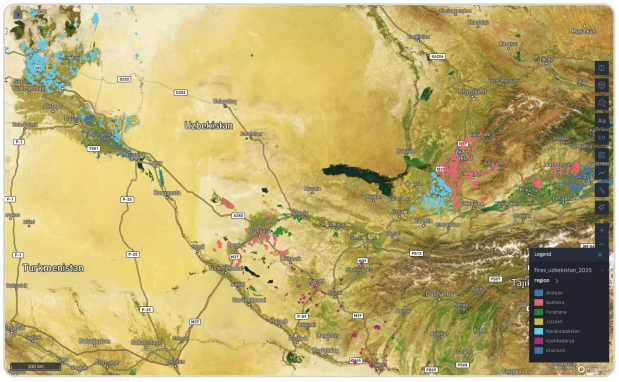
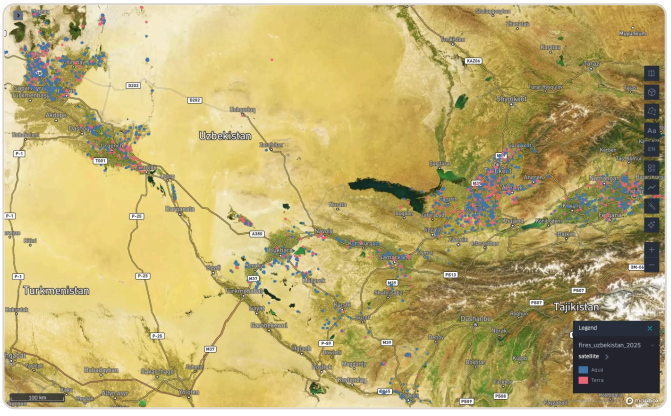
🎨 Part 6 – Make it yours

Everyone in the group uses the same data. Here's how to make **your** map different. Pick **one idea from each row** (or invent your own!):

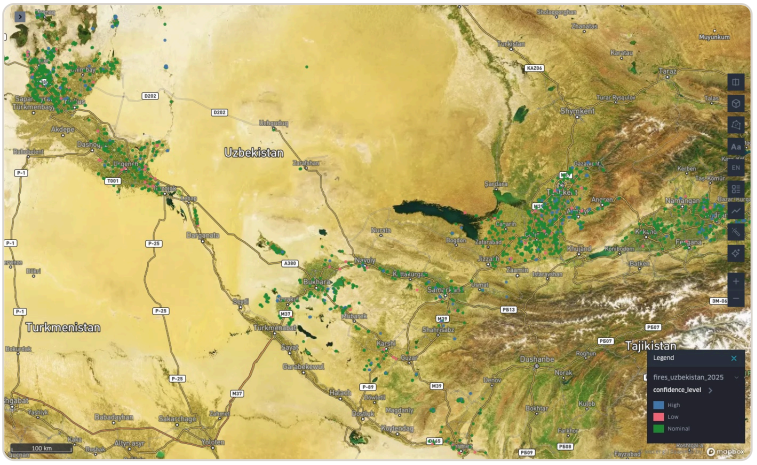
	Choose one
1. What to color by (Step 10)	`day_night` · `season` · `region` · `satellite` · `confidence_level` · `fire_type` · `month`
2. Where to look	All of Uzbekistan · your home region · the Fergana Valley · Karakalpakstan & the Aral Sea · a single city
3. Background (Step 6)	Satellite · Dark · Light · Voyager · No basemap

Here's what some of the combinations look like:

A · Day or night? (`day_night`)	**B · Which season? (`season`)**
	
About 1 in 3 detections happened at night.	**Spring** fires dominate the farmland along the Amu Darya in Khorezm and Karakalpakstan.

C · Which region? (`region`)	**D · Which satellite? (`satellite`)**
	
Set **Steps** → **12** so each region gets its own color.	Aqua (afternoon pass) caught 2,679 points, Terra (morning pass) 1,970.

E · How sure is NASA? (`confidence_level`)



More ideas to stand out ⭐

- **Size by strength:** in the layer settings, open **Radius** and set **Radius Based On** to `fire_power_mw`. Strong fires become big circles.
- **Only one month or season:** open the **filter tab** (the funnel icon, 2nd tab) → **Add Filter** → choose `month` or `season`, then pick the values you want.
- **Fires as a heatmap:** under **Basic**, change *Point* to **Heatmap** for a glowing "density" look.
- 🕵️ **Solve the mystery:** about **1,700 hotspots** are stacked on almost the same spot near **38.05° N, 67.30° E** in Surkhandarya. They all look like one dot! Zoom in there on the satellite background. What do you think is burning all year round?
- **Compare years:** download another year from NASA (**Option B**), e.g. 2015 vs 2025, and load both files. Each becomes its own layer.

📤 Part 7 – Share your map

1. Make a folder named after **your map's topic**, e.g. `Fires by season`. Put your image inside, plus a short `README.md` with **your name**, the data and the tools you used.
2. Upload the whole folder to **`2026/Day 01 Points/`** in this repository. The Quick Start in the main README shows how, with no coding needed.
3. Post it with **#30DayMapChallenge** and say which choices you made (e.g. "*colored by season, Fergana Valley, dark background*").

🆘 Something went wrong?

Problem	Fix
The **Add Data** window disappeared	Click **+ Add Data** at the top of the left panel.
Nothing happens after pasting the link	Check that you copied the **whole** link, or download the file and drag it in instead (Part 1).
I can't find the downloaded file	Look in your **Downloads** folder. In Chrome, press **Ctrl + J** (Cmd + Shift + J on a Mac) to see your downloads.
All points are the same color	Make sure you chose a column in **Color Based On** (Step 10).
Colors look too similar	Use a **Qualitative** palette (Steps 11–13).
The map is blank or gray	Wait a few seconds for the background to load, or zoom in and out once.
I closed the tab and lost everything	Nothing is saved online. Just drag your file in again (Step 2); it only takes a minute to get back.

📚 Data credits

- **Fire detections:** NASA FIRMS, MODIS Collection 6.1 active fire / hotspot detections (MCD14ML), Terra & Aqua, Uzbekistan 2025. firms.modaps.eosdis.nasa.gov. doi: 10.5067/FIRMS/MODIS/MCD14ML
- **Region names:** added using the region boundaries in `2025/Boundaries` (source: simplemaps.com). Points just outside the border were matched to the nearest region.
- **Readable columns** (`fire_type` , `day_night` , `season` , `month` , `confidence_level`) were made from the original FIRMS codes: `type` , `daynight` , `acq_date` and `confidence` (Low < 30, Nominal 30–80, High > 80).